graduate students and should be accessible to students with a strong background in linear algebra and a basic knowledge of abstract algebra. Theoretical material in this book is supplemented by many problems and exercises which touch upon a lot of additional topics; the more difficult exercises are provided with hints.

The book covers a number of standard topics in representation theory of groups, associative algebras, Lie algebras, and quivers. For a more detailed treatment of these topics, we refer the reader to the textbooks [**S**], [**FH**], and [**CR**]. We mostly follow [**FH**], with the exception of the sections discussing quivers, which follow [**BGP**], and the sections on homological algebra and finite dimensional algebras, for which we recommend [**W**] and [**CR**], respectively.

The organization of the book is as follows.

Chapter 2 is devoted to the basics of representation theory. Here we review the basics of abstract algebra (groups, rings, modules, ideals, tensor products, symmetric and exterior powers, etc.), as well as give the main definitions of representation theory and discuss the objects whose representations we will study (associative algebras, groups, quivers, and Lie algebras).

Chapter 3 introduces the main general results about representations of associative algebras (the density theorem, the Jordan-Hölder theorem, the Krull-Schmidt theorem, and the structure theorem for finite dimensional algebras).

In Chapter 4 we discuss the basic results about representations of finite groups. Here we prove Maschke's theorem and the orthogonality of characters and matrix elements and compute character tables and tensor product multiplicities for the simplest finite groups. We also discuss the Frobenius determinant, which was a starting point for development of the representation theory of finite groups.

We continue to study representations of finite groups in Chapter 5, treating more advanced and special topics, such as the Frobenius-Schur indicator, the Frobenius divisibility theorem, the Burnside theorem, the Frobenius formula for the character of an induced representation, representations of the symmetric group and the general